LMAP — Liquid Media Access Protocol

A Peer-to-Peer Protocol for Media Distribution and Ownership

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Abstract

A purely peer-to-peer system for film distribution would allow digital media ownership to exist independently of any single platform, studio, or streaming service. The current model of centralized streaming creates fundamental misalignments: platforms bear infrastructure costs proportional to viewership, creators receive opaque compensation, and consumers own nothing permanent. We propose a layered protocol in which each film exists as fungible tokens within a unified registry contract, with separate provision for copyright ownership. Token quantity determines usage rights — from personal viewing to theatrical exhibition — at thresholds configured per title. Encrypted content is stored on a content-addressed network. Two attestation tiers coexist within the protocol: an open tier that uses deterministically-derivable keys for permissive content, intentionally permeable to preserve platform-independent ownership; and a certified tier that uses hardware-attested per-device key wrapping for content where contractual digital-rights management is required. A network of user-operated nodes — Seeds — distributes content, inverting the conventional cost model: the network strengthens as viewership grows. Smart contracts automate royalty distribution on every transaction, including secondary sales where collectors can sell exhibition rights directly. The protocol is governed by an independent foundation, structurally distinct from any commercial operator. We argue that liquid media — physical-grade ownership in liquid digital form — is achievable without compromising the sovereignty that makes the design worthwhile.

1. Introduction

The transition to streaming was marketed as disruption, but the underlying economics reveal a fundamentally broken model. When a film succeeds on a centralized platform, infrastructure costs increase proportionally — the platform is economically punished for having popular content. This creates incentives misaligned with the goal of connecting audiences with compelling stories.

The consumer experience has equally degraded. Digital purchases are platform-locked, creating fragmented collections scattered across services. When platforms sunset or lose licensing rights, purchased content disappears. Unlike physical media, there is no true ownership — only temporary access privileges revocable at the platform's discretion.

Creators face similar opacity. Residual payments flow through accounting systems designed to obscure rather than illuminate. The shift from theatrical and physical media sales to streaming has compressed compensation without providing comparable transparency or participation in secondary markets.

What is needed is a protocol-level solution addressing all constituencies simultaneously — analogous to how DVD or USB function: not as a platform a consumer signs up for, but as a *standard* that many implementations interoperate through. We propose LMAP (the Liquid Media Access Protocol): a protocol where digital media is

tokenized into standardized units granting verifiable ownership and access rights; content is distributed through a peer-to-peer network that scales positively with demand; smart contracts ensure automatic royalty distribution on every transaction; and ownership is structurally separable from any particular platform that runs on the protocol.

What consumers experience is *liquid media* — a generic public term for the category LMAP enables. Liquid media is to streaming services what physical media was to broadcast television: a different form of ownership, with different rights, that buyers can hold, transfer, and pass on.

2. Token Architecture

Each film exists as a unique token ID within a master ERC-1155 [1] registry contract. Unlike ERC-721 [2] non-fungible tokens, where each token is distinct, ERC-1155 allows millions of identical tokens per film, enabling both mass distribution and liquidity. A single contract scales to thousands of films, each identified by a unique token ID, with configurable quantities per title.

The token's URI points to a content identifier on a distributed storage network containing immutable metadata: title, synopsis, credits, and references to media assets. This architecture ensures that even if any particular implementation of the protocol ceases operations, token holders retain permanent access through the decentralized network and the open protocol specification.

Implementation status: shipped on Polygon mainnet as the `WyllohRegistryProtocolV4_1` contract. Verified source available via public block explorers.

3. Separation of Distribution and Copyright

A critical distinction exists between distribution rights and underlying copyright. The ERC-1155 registry manages distribution tokens — fungible units representing viewing and exhibition rights. A separate ERC-721 contract is specified to establish copyright ownership: a single non-fungible token per film representing the intellectual property itself.

The copyright token grants control over derivative works, merchandising, and the ability to propose governance actions such as token splits. Distribution token holders own rights to view, exhibit, and trade — but not to create sequels or merchandise. Crucially, the copyright holder cannot unilaterally mint additional distribution tokens; supply modifications require token holder approval, protecting holders from dilution. This mirrors traditional film finance, where investors hold revenue participation without controlling the underlying intellectual property, while adding protections absent in conventional structures.

This separation enables powerful scenarios. A studio could acquire the copyright token to produce a sequel while existing distribution tokens continue trading. The original filmmaker benefits from the sale; distribution token holders retain their viewing rights and market participation. Rights flow to natural owners without forcing all-or-nothing transactions.

Implementation status: design intent. The current V4.1 registry is the ERC-1155 distribution layer; the ERC-721 copyright registry is specified for a future protocol version (V5+) and has not yet been deployed. The current protocol's role-gated minting (see §15) serves as a near-term substitute for the copyright registry's supply-modification controls.

4. Modular Rights Through Stacking

Token quantity determines usage rights through a threshold-based licensing system. A single token grants personal viewing rights — analogous to owning physical media. As tokens accumulate in a wallet, additional rights unlock at thresholds the filmmaker configures at tokenization time.

Thresholds are intentionally not fixed by the protocol; each filmmaker sets thresholds appropriate to their content's economics. A documentary might set low thresholds to encourage screenings; a genre release might set high thresholds reflecting commercial value. The protocol does not impose a universal price for theatrical rights because no such universal price exists in reality. The market discovers appropriate pricing through trading at the thresholds the filmmaker has chosen.

For illustrative purposes only — not as protocol-level constants — typical configurations might include:

- *1 token* — personal viewing rights
- *100 tokens* — small-venue exhibition
- *60,000 tokens* — theatrical exhibition rights with access to professional distribution formats

But these are choices, not specifications. Thresholds for any given film are encoded in that film's metadata at minting and are immutable thereafter for the life of that token's contract.

This mechanism creates natural price discovery for commercial rights. Rather than negotiating complex licensing agreements, distributors acquire sufficient tokens on the open market. Token supply is fixed at initial minting; if market conditions warrant increased liquidity, token holders may vote to approve a split, multiplying token quantities while preserving proportional ownership. The copyright holder may propose splits but cannot execute them unilaterally. New creative works — sequels, adaptations, alternative cuts — are tokenized as separate token IDs with independent supplies, leaving original token markets undisturbed.

Implementation status: shipped. Stacking thresholds and on-chain enforcement are operational in V4.1.

5. Layered Architecture

The protocol decomposes into eight layers. Each layer has a defined responsibility, a published interface to the layers above and below it, and an implementation status. Different parties can build different layers as long as the interfaces are honored. Conflating layers — particularly the entitlement layer with the cryptographic layer — is how comparable projects have failed.

#	Layer	Responsibility
0	Trust Anchors	Cryptographic primitives; hardware roots; wallet identity; contract immutability
1	Storage	Content distribution substrate (IPFS, manifests)
2	Cryptography	Encryption, key hierarchy, dispersal, watermarking
3	Entitlement	Smart contracts: distribution + copyright registries, royalties, rights tiers
4	Network	Peer discovery, shard request/response, LAN streaming
5	Attestation	Open tier / certified tier hardware attestation
6	Application	Storefronts, library UI, playback clients, integrator tooling
7	Governance	Foundation structure; federated certification; protocol governance

A storefront (Layer 6) only needs to read entitlement (Layer 3); a Seed manufacturer needs to implement Layers 1, 2, 4, and optionally 5; an academic auditor of the certified tier only needs to engage with Layer 5 and the cryptographic claims of Layer 2. Each layer should be specifiable, testable, and replaceable independently of the others. The protocol does not mandate that any single party implement all eight layers.

The remaining sections of this paper traverse these layers from storage upward.

6. Content Storage

Media files are encrypted using AES-256-GCM before distribution on a content-addressed storage network [3]. The metadata structure nests content identifiers: the token URI points to a metadata file containing references to both unencrypted promotional assets (poster, trailer) and encrypted media files. Each encrypted asset specifies a threshold — the token quantity required for access — corresponding to the rights tiers established at minting.

Encrypted content is stored in a chunked format (`[Length(4 bytes)][IV(12 bytes)][Ciphertext+Tag(16 bytes)]` per 4-MiB chunk) which is self-authenticating per chunk and supports streaming decryption with constant memory. The format is documented in the protocol specification; any compliant client can implement encryption and decryption against it.

A future enhancement at this layer is *threshold dispersal of ciphertext shards* — distributing encrypted content shards across the Seed network using Reed-Solomon erasure coding, with the option of secret-share-based reconstruction [4], such that reconstruction requires a quorum of nodes. This is a resilience property at the storage layer, not the load-bearing security mechanism (see §7.4 for the precise scope of this claim).

Implementation status: chunked AES-256-GCM encryption and IPFS storage are shipped. Threshold ciphertext dispersal is specified for a future protocol version.

7. Cryptography and the Two-Tier Attestation Model

The protocol resolves the open-protocol-versus-studio-trust tension at Layer 5 (attestation), not at Layer 2 (encryption). Two tiers run the same protocol underneath; the gating between them is narrow and hardware-attested.

7.1 Open Tier

Any compliant Seed implementation serves content licensed under permissive terms — public domain, Creative Commons, indie creators who opt their content into open-tier distribution.

In the open tier, a film's master content key is derivable from public on-chain data using a documented formula:

The encrypted master key, in turn, is delivered to a buyer by a storage service after that service verifies on-chain that the buyer's wallet holds at least one token. Once the buyer has the encrypted master key, derivation of the decrypting key is purely local and does not depend on any external service.

This permeability is a deliberate design choice. **A developer with the public protocol specification could write a fifty-line program that downloads and decrypts an LMAP-tokenized film without ever touching any Wylloh-operated server.** The open tier accepts this and makes it explicit, because the alternative — wallet-bound or service-dependent decryption — would break the platform-independence guarantee that gives the protocol its long-term value.

In open-tier security terms, this is approximately DVD-level protection. A determined party with technical knowledge can extract plaintext; the casual sharing case ("burn a copy and hand it to a friend") does not extract anything that the friend could not have purchased themselves for $4.99 USDC. Defense rests on social incentive, the ease of legitimate purchase, and the alignment of buyers with creators — not on cryptographic enforcement against sophisticated extraction.

7.2 Certified Tier

Studios and rights holders who require contractual digital-rights management cannot use the open tier as-is. The certified tier provides hardware-attested per-device key wrapping for content where this is required.

In the certified tier:

- Each certified Seed holds a per-device keypair generated inside its hardware secure element. The private key is non-extractable.
- Content keys are wrapped to each certified Seed's public key. Compromising one certified Seed yields *that Seed's local content only*; the network's aggregate exposure does not grow with the network's size.
- Wrapped keys are unwrapped only inside the secure element. The unwrapped key never appears in main memory or on persistent storage outside the secure perimeter.
- Issuance of a wrapped key requires the Seed to present a fresh *attestation report* signed by its secure element, containing current firmware measurements. The report is verified against the Seed's attestation credential before any wrapped key is issued.

Reference hardware for the certified tier includes ARM TrustZone [5]-capable SoCs with discrete secure elements (e.g., NXP SE050 or equivalent), or platforms with TPM 2.0 [6] support. Remote attestation [7] follows established patterns from contemporary trusted-computing literature. Per-title content keys are derived from a master with HKDF-style domain-separated key derivation [8] to prevent key reuse across titles.

Hardware binding is a normative requirement of the certified tier. A compliant certified-tier implementation MUST use a hardware secure element to (a) generate the per-device keypair, (b) hold the private key in non-extractable form, (c) perform key unwrapping, and (d) sign attestation reports. An implementation that performs any of these operations outside a hardware secure element is not conformant with LMAP V6 and MUST NOT be marketed as such. This is the protocol-level guarantee studios rely on when authorizing content for the certified tier. Wire-level specifics of the attestation report format, the attestation credential schema, and the issuer federation protocol are committed to a separate LMAP V6 technical specification, forthcoming.

A clarifying distinction: only the *decryption* step requires certified hardware. Encrypted certified-tier content is freely distributable through the same content-addressed substrate as open-tier content — any device can hold, pin, or serve it. Tokens remain freely transferable across all wallets. The certified tier gates *playback*, not *carriage*. This preserves user sovereignty (a buyer's ownership is unaffected by hardware availability; they can transfer their token at any time) while satisfying the studio requirement (decryption is impossible without an attested secure element). The architectural model mirrors theatrical distribution under DCI: the encrypted DCP is portable and freely distributable; only certified projectors with KDM-bound keys can play.

7.3 Attestation Issuers and the Federated Framework

Attestation credentials in the certified tier are signed by independent *attestation issuers*. An issuer is a party that verifies a Seed's hardware integrity and signs a credential binding the device's public key to a manifest of approved firmware measurements. Issuers in turn operate under a federated *attestation framework* governed by the Liquid Media Foundation (see §15).

No single issuer can unilaterally approve or deny attestation; the framework's governance can revoke any issuer whose practices fail audit. Studios negotiate with the framework, not with a single platform operator. The model is structurally analogous to the federation of certificate authorities that underpins the public web's TLS infrastructure: independent issuers, audited practices, root trust managed by a body distinct from any commercial participant.

The framework launches with a single issuer (the Liquid Media Foundation itself) and a public commitment to federate at a defined milestone of network maturity. Federation is therefore a visible commitment, not an indefinite promise.

7.4 Security That Scales with Network Size

Two distinct claims about how protocol security relates to network size deserve separate treatment, because they have different defensibility.

Strong claim — per-device key wrapping (certified tier). Compromising one certified Seed yields that Seed's local content only. The network's aggregate exposure does not grow with the network's size, because each certified device is its own isolated vault. This is the load-bearing security claim, and it scales positively with the number of Seeds in the network: more participants do not increase the attack surface against any single film.

Modest claim — threshold dispersal of ciphertext (storage layer). Sharding ciphertext across Seeds raises the per-shard extraction cost. An attacker compromising one Seed obtains a fraction of the ciphertext for any given film; reconstruction requires reaching the quorum threshold. This is real but bounded: a sufficiently motivated attacker can target enough Seeds to reach quorum. The combinatorial cost is meaningful but does not produce asymptotic security.

The protocol therefore claims that *per-device key wrapping at the certified tier provides security that does not degrade with scale*, and that *threshold dispersal provides resilience and modestly raises extraction cost*. Both are valuable; only the first is the load-bearing claim.

This precision matters for technical readers because the conventional DRM model degrades as user base grows (more devices, more keys, more attack surface). The certified tier inverts that property; the storage layer's dispersal supplements it.

7.5 Watermarking

Forensic watermarking binds a leak to a specific wallet, raising the cost of unauthorized distribution beyond the bare cryptographic extraction cost. In the open tier, a watermark may be inserted server-side by a storage service when fulfilling a download. In the certified tier, watermarks are inserted Seed-side at decryption time, inside the secure element, before plaintext leaves the secure perimeter.

Watermark robustness against transcoding and adversarial filtering is an ongoing research area; commercial watermarking implementations exist with established studio relationships. The protocol specifications watermark insertion points but does not mandate a particular scheme.

Implementation status: open tier (deterministic key derivation, chunked AES-256-GCM, self-contained manifest reading from on-chain metadata + IPFS, with storage-API fallback for tokens whose metadata predates the self-contained format) is shipped and demonstrated end-to-end on independent hardware. Certified tier (per-device key wrapping, hardware attestation, federated issuers) is specified at the protocol level with hardware-binding declared normative (§7.2); the wire-level attestation specification, the framework's first issuer, governance process, and operational maturity are forthcoming.

8. Permanence Guarantees

The protocol commits to a permanence guarantee in both tiers: a Seed that has legitimately acquired a license to a film at any point in time retains the ability to play that film indefinitely, regardless of subsequent network availability, attestation revocation, or platform fate.

In the open tier, this guarantee is structural. The decryption key for any owned film is derivable from public on-chain data using the documented formula. A Seed unplugged for twenty years and plugged back in can play any film it previously legitimately acquired, even if the Liquid Media Foundation has ceased to exist, with zero network connectivity, using a reference client a third party authored from the public protocol specification. The blockchain remembers what was bought; nothing else needs to.

In the certified tier, this guarantee is engineered. Attestation gates the *issuance* of wrapped keys, not the *playback* of content already keyed. A Seed whose attestation credential is later revoked retains the wrapped keys it previously received and continues to play the corresponding films. Revocation prevents the Seed from acquiring further content through the certified tier; it does not retroactively disable the Seed's existing library. This is the architectural answer to the rugpull pattern that has characterized streaming services and digital media stores.

The hardware itself remains a single point of failure: a Seed whose secure element fails physically loses access to its locally-wrapped keys. The protocol mitigates this by treating the on-chain ownership record as the durable source of truth. A user whose Seed has died can re-acquire wrapped keys for their owned films onto a new certified Seed, because the wallet's holdings are recoverable from the blockchain. **The blockchain remembers; the hardware is replaceable.** This is a stronger guarantee than physical media, which has no equivalent recovery path when a disc fails.

9. The Seed Network

Content distribution operates through *Seeds* — user-operated nodes that combine encrypted-content storage, peer-to-peer participation, and a local LAN-streaming service for playback applications. Seeds run on commodity hardware following published reference designs.

A Seed is a headless device. It does not connect directly to a television; instead, it serves LMAP-registered content to client applications running on platforms users already own — Roku, Apple TV, iOS, the web. The user's existing remote, the user's existing television, the user's existing input habits all remain intact. The Seed becomes infrastructure: quiet, reliable, operated as a piece of furniture in the home network rather than a screen in the living room. Earlier protocol designs envisioned the Seed as a custom-UI device that attached directly to a television; experience has shown that a headless server paired with thin client applications on existing platforms is simpler, more accessible, and less subject to the obsolescence that affects custom television-UI hardware.

Each Seed stores content its operator owns, plus optionally contributes overflow capacity to the network. Popular films accumulate across more Seeds as more users own tokens; niche content maintains availability proportional to its audience. No central coordination determines what Seeds store; ownership patterns drive distribution.

This architecture inverts conventional streaming economics. Centralized platforms bear infrastructure costs proportional to viewership: success is expensive. In the Seed network, increased viewership means more Seeds joining, distributing load across more nodes. Per-stream costs approach zero as the network grows. The system becomes more efficient at scale rather than more costly. Network participation may be incentivized through future token mechanisms (proof-of-storage rewards, bandwidth credits); in the near term, Seeds participate by default because their primary value to the operator is local — the films they own, accessible to LMAP-compatible playback clients on their local network — and network participation is a beneficial side effect rather than the purchase rationale.

A reference Seed device — the *Origin* — is specified separately from this paper as a buildable open-hardware design. The protocol does not require the Origin or any particular Seed implementation; any compliant device can participate.

Implementation status: an open-source reference Seed daemon is operational. It verifies wallet ownership directly against the on-chain registry, fetches encrypted content from IPFS (preferring a co-resident Kubo node where present, falling back to public gateways), decrypts streaming with constant memory, automatically pins the source CID to make the Seed a provider for the content it holds, and exposes a long-running HTTP service with mDNS auto-discovery and HTTP byte-range streaming for LAN client applications. End-to-end operation has been demonstrated on independent ARM64 hardware (Raspberry Pi 4), with byte-identical decrypted output across architectures and with no dependency on any centralized service. The reference *Origin* device is in pre-production. The web client at wylloh.com serves as the first reference application.

10. Marketplace Mechanics

The protocol includes integrated marketplace functionality. Primary sales occur through minting at prices set by the title's author. Secondary sales enable peer-to-peer trading with automatic royalty distribution on each transaction.

10.1 Three-Way Revenue Split

Every primary sale routed through the protocol distributes payment across three parties, with proportions set per title at mint time:

- **Protocol fee: 2.5%** (immutable, hardcoded at the contract level). Flows to the Liquid Media Foundation treasury for shared infrastructure: IPFS pinning redundancy, contract auditing, security work, open-source maintenance.
- **Publisher fee: 0–25%** (set per title, immutable for that title). Flows to the publisher — the wallet that minted the title. Publishers are entities that bring works to market: established storefronts, indie collectives, or self-publishing authors. Their cut compensates curation, marketing, audience development, and brand value.
- **Author share: remainder.** Flows to the author — the credited creator(s). Distributed via an optional royalty shareholders configuration (§10.3) supporting up to 50 collaborators per title.

For a typical configuration — 10% publisher fee — the split is 2.5% / 10% / 87.5%. Authors receive dramatically better economics than traditional publishers (where author shares of 10–20% are common). The protocol's 25% publisher cap is the only opinionated stance: *publishing on LMAP shall not become as extractive as the systems it replaces*.

10.2 Publisher and Author as Distinct Roles

The protocol records two distinct identities per title:

- **Publisher** is set automatically from the wallet that calls the mint function. Cryptographically authentic — cannot be impersonated. Establishes the brand identity associated with the title and receives the publisher fee on primary sales.
- **Author** is provided at mint and identifies the credited creator of the work. Receives the author share of every primary and secondary sale.

The two roles are often the same wallet (an author who self-publishes) but can differ (a publisher mints under their brand and routes the author share to a separate creator wallet). This mirrors the publishing-house / author distinction familiar from books, films, and music.

10.3 Royalty Shareholders

Authors can configure up to 50 royalty shareholders per title, each receiving a configurable percentage of the author's share. This supports the realistic case where a tokenized work has multiple contributors deserving compensation: a film with director, lead actors, composer, screenwriter; a video game with developers, artists, and key team members; an album with songwriter, performers, and producer.

Shareholders are mutable by the author, allowing collaborators to be added or adjusted as the work evolves. Sum of shares can be ≤ 100% of the author's share; any remainder flows to the author wallet, preserving the author as the residual recipient. For projects with more than 50 beneficiaries, shareholders compose with external splitter contracts (e.g., 0xSplits): one shareholder slot can route to a splitter address that distributes further.

Distribution happens automatically on every protocol-routed sale — both primary and secondary. No manual royalty disbursement, no escrow, no off-chain ledger reconciliation.

10.4 Royalty Enforcement Boundary

The protocol enforces royalties cryptographically for sales routed through its marketplace functions. External marketplaces honoring the ERC-2981 royalty standard will respect the configured royalty when displaying and processing sales. Direct wallet-to-wallet transfers (the gift, inheritance, and personal-give use cases) do not trigger royalties — this is a property of the underlying ERC-1155 standard and a deliberate preservation of token-as-property semantics.

Off-chain payment with on-chain gift is theoretically possible and unenforceable cryptographically, as with every royalty-bearing token system. The protocol's mitigation is structural: the low 2.5% protocol fee minimizes the savings from circumvention; the integrated marketplace makes legitimate sales easier than coordination; on-chain transparency makes wash-trade patterns visible and reputationally costly.

10.5 Bulk Acquisition for Rights Stacking

The marketplace supports atomic batch purchases (`batchBuyListings`), enabling a single transaction to acquire tokens from many sellers simultaneously. This is essential for the rights-stacking use case (§11): a buyer assembling tokens to meet a theatrical-exhibition threshold may need to acquire thousands of tokens from hundreds of sellers in a single coordinated purchase. The atomic semantics guarantee all-or-nothing: if any listing in the batch fails, the entire transaction reverts, preventing the partial-fill case where a buyer pays fees but doesn't reach their target threshold.

Platform UIs aggregate, sort, and present available listings; the protocol provides the batch-execution primitive.

10.6 The Protocol Fee Is Not a Platform Fee

The 2.5% protocol fee supports shared infrastructure all implementations depend on, identical across every LMAP-compatible marketplace. Third-party marketplaces may layer their own commercial markup on top, competing on user experience, curation, and discovery rather than by undercutting the protocol fee. This places third-party marketplaces and the reference web client at wylloh.com in a peer relationship with one another, not a competitive one.

10.7 Settlement and Gas Abstraction

Transactions settle in stablecoin to insulate buyers from cryptocurrency volatility. The current implementation uses USDC.e (bridged USDC) on Polygon. Settlement on Polygon means that buyers must currently hold both USDC.e and a small amount of MATIC to cover network gas — friction the protocol acknowledges as a near-term limitation rather than an intended feature. The architectural answer is account abstraction via ERC-4337[9] *paymaster contracts*: a paymaster pays MATIC gas on the user's behalf, recovering the cost via a small USDC surcharge bundled into the transaction. The reference contract V5 exposes paymaster hooks allowing a paymaster contract to be wired in by foundation governance. Paymasters are deployed and funded at the marketplace layer; each marketplace can choose whether and how to subsidize gas as part of its commercial offering.

Implementation status: primary sales with three-way distribution, secondary sales with royalty enforcement, royalty shareholder distribution, batch acquisition, and the protocol fee are specified in the V5 reference contract. The V5 contract is in active development; deployment under foundation governance is a 2026 milestone. The legacy V4.1 registry remains in operation for existing tokens (open-tier only, two-way author/protocol split) during the transition.

11. Commercial Rights Discovery

The stacking mechanism enables a novel form of commercial rights acquisition. Consider a film that achieves cult status: tokens are distributed across thousands of fans who purchased during initial release. A theater chain wishes to conduct a theatrical run, requiring tokens above the exhibition threshold the filmmaker configured for that title.

Rather than negotiating with the original filmmaker, the theater acquires tokens directly from fans through open market purchases. As buy pressure increases, token prices rise, rewarding early believers who funded the film's success. Royalties trigger on each transfer, flowing to the copyright holder regardless of who sells. The market determines that theatrical rights to this particular film, at this particular moment, command a specific price — no negotiation required.

Commercial exhibition requires tokens to be staked for the exhibition period. The professional distribution system verifies on-chain that staked tokens meet the exhibition threshold before enabling playback. This prevents the exploitation pattern of acquiring tokens, screening content, and immediately selling. Staking creates commitment aligned with legitimate commercial use.

Implementation status: design intent. Stacking thresholds are operational at the contract level today; the staking mechanism for commercial-exhibition windows is specified for a future protocol version.

12. Production Funding

The token architecture enables presale-based production financing. Rather than relying on studio executives or platform algorithms to greenlight projects, creators can presell tokens directly to audiences. This validates market demand before production begins while distributing ownership of the work directly to its audience.

Presale tokens grant the same ownership rights as post-release tokens but are purchased at a discount, rewarding early supporters. Smart contracts can structure milestone-based fund releases with multi-signature approval from stakeholder representatives. If production fails to meet milestones, remaining funds return to presale participants.

This model removes centralized gatekeepers from greenlighting decisions. A film need not fit a platform's content strategy. If sufficient audience interest exists to fund production, the project proceeds. The presale serves simultaneously as market research, financing mechanism, and community-building exercise.

Implementation status: design intent. The token architecture supports presales architecturally; the milestone-based escrow contracts and the corresponding presale-tokenization workflow are specified for a future protocol version.

13. Privacy

Contemporary streaming platforms extract extensive personal data to drive recommendation algorithms and advertising. The protocol operates on a different principle: *movies should not watch their viewers*. Analytics derive exclusively from public blockchain data — transaction patterns, token distribution, market activity — rather than viewing behavior or personal information.

The conventional streaming model links user identity to viewing history to platform to advertisers, forming a chain of surveillance maintained primarily for monetization through inferred consumer preferences. The protocol model maintains pseudonymous wallets executing public transactions on a transparent ledger, with no link to real-world identity, no telemetry from playback events, and no viewing patterns aggregated by any party. Creators receive aggregate insights — token distribution by region inferred from public data, secondary-market activity, holder concentration — without compromising individual privacy.

14. Security Philosophy

The protocol does not attempt perfect copy protection. History demonstrates that technological restrictions on digital media inevitably fail while degrading legitimate user experience. Instead, the system aligns incentives such that ownership provides value beyond mere access — liquidity, permanence, community membership — making legitimate acquisition the rational choice.

The analog hole — capture of content via camera or screen recording during legitimate playback — remains possible. Watermarking is the only mitigation, and watermark robustness against transcoding is an ongoing arms race. The protocol does not claim immunity from this class of attack; it claims that legitimate purchase confers economic benefits that bare ciphertext extraction does not.

When content is owned not by a distant corporation but by potentially many individual token holders, piracy dynamics shift. Unauthorized distribution harms not an abstract studio but a community of fellow enthusiasts. Token holders own copies of the work itself, aligning their interest with its integrity rather than its piracy. Social contracts, properly constructed, provide stronger protection than technological locks against the casual case; the certified tier provides cryptographic protection against the sophisticated case where contractual DRM is required.

Physical media operated on similar principles. Copy protection on optical discs was trivially circumvented, yet the market thrived for decades. Convenience, quality assurance, and social norms around ownership provided sufficient protection. The protocol adds financial incentives physical media lacked: ownership appreciates in genuinely successful work, and secondary markets reward early adoption.

A specific guidance for technical readers: when evaluating the protocol's claims, separate the *cryptographic claim* (the certified tier's per-device key wrapping does not degrade with network scale) from the *operational claim* (legitimate purchase is more economically rational than extraction for participants in the network). Both claims hold, but they hold at different layers and under different threat models. The whitepaper is precise about this because the audiences this document serves — investors, developers, and studio business affairs — read the same words and need to extract the same meaning.

15. Governance and the Liquid Media Foundation

The protocol is governed by an independent foundation, structurally distinct from any commercial operator. The foundation holds the protocol's intellectual property under a permissive open-source license, stewards the certified-tier attestation framework, publishes the protocol specification, and operates as a protocol-neutral standards body.

Commercial entities that build on the protocol — storefronts, hardware manufacturers, marketplace operators — are participants in the LMAP ecosystem rather than its operators. This structural separation mirrors precedents in adjacent ecosystems: the Filecoin protocol governed by the Filecoin Foundation, structurally distinct from Protocol Labs; the Ethereum protocol governed independently from any single Ethereum company. The pattern preserves credible-neutrality: filmmakers, third-party storefronts, hardware manufacturers, and eventually studios can build on the protocol without fearing that the protocol's stewards are also their commercial competitors.

The foundation, at its mature governance, will steward:

- **Specification governance.** Protocol upgrades, EIP-equivalent proposals, and version transitions.
- **The attestation framework.** Maintaining the federation of attestation issuers, audit standards, and revocation governance for the certified tier (§7.3).
- **Treasury.** Custody of the protocol fee revenue, application to shared infrastructure costs, and grants to ecosystem development.
- **Token-holder governance.** Per-title decisions (royalty splits, rights-tier amendments, supply changes) where applicable.

A pragmatic note about the present: the foundation does not yet exist as a formal legal entity. The current governance posture is that of a stewarded bootstrap, in which the founding team operates the protocol on the implicit promise of formalizing the foundation and transitioning to community governance over a multi-year arc. This is a commitment, not a fact, and the protocol's roadmap includes the foundation's formal establishment, the attestation framework's first audit milestones, and the public sunset of any admin keys held by the founding team.

Curation: gateless, not absent. The protocol layer is permissionless — any wallet can mint a title (V5 onward). Curation happens at the application and registry layers, not at the protocol. Three signals work together:

1. **Cryptographic identity at mint.** The wallet that mints a title is recorded as the publisher, on-chain, immutably. Brand identity is wallet-bound and cannot be impersonated.
2. **Foundation-maintained verified publishers registry.** The Liquid Media Foundation publishes a curated list of verified publishers at `lmap.org/publishers.json`. Verification signals quality and accountability without restricting the protocol; unverified publishers can still mint, but won't appear in verified-only application views.
3. **Application-layer editorial choice.** Each platform built on LMAP applies its own filter: the Wylloh storefront shows titles from Wylloh and other respected publishers; an indie aggregator might show all unverified content; a music-focused platform filters by `mediaType: music` and music-publisher wallets. The protocol exposes the truth (who minted what); applications shape the experience.

This model — *permissionless protocol, curated applications* — mirrors how DNS works: anyone can register a domain, but resolvers, browsers, and search engines each apply their own filters. Quality emerges economically rather than by gatekeeping. A publisher with poor taste mints work nobody buys; a publisher with good taste accumulates verified status and earns the publisher fee from real demand.

The legacy V4.1 registry retains its `FILM_CREATOR_ROLE` gate as the editorial mechanism for the open-tier-only deployment serving already-minted titles. New tokenizations under V5 land in the permissionless model.

Implementation status: the foundation is in formation (incorporation as a 501(c)(6) trade association is a funded milestone of the upcoming raise); V5's permissionless minting is specified and implemented in the reference contract scaffold; deployment under foundation governance is a 2026 milestone; the verified publishers registry will launch concurrently with V5 deployment.

16. Comparison with Prior Work

LMAP inherits ideas from several adjacent ecosystems and addresses limitations specific to each.

Centralized streaming services (Netflix, Disney+, et al.) own the relationship with audiences and bear infrastructure costs proportional to viewership. They cannot offer durable digital ownership because durable ownership conflicts with subscription economics. LMAP inverts this by routing ownership through a public ledger and distribution through a peer-to-peer network.

Kaleidescape ships hardware-backed ownership for theatrical- quality content, but operates within a closed ecosystem at luxury price points and serves a narrow demographic. LMAP shares the philosophical commitment to genuine ownership but extends the addressable audience by separating the protocol from any particular hardware vendor and accepting standard 4K HDR streaming quality rather than requiring lossless masters.

Plex demonstrates the "headless server at home plus client app on the television" architecture at scale (over 25 million users) with primarily user-ripped content of unclear provenance. LMAP- compatible Seed devices inherit this architecture and address Plex's structural limitations: no provenance, no royalty flow, no curatorial focus, closed-source business model. The reference Seed device extends Plex's pattern with on-chain provenance and an open protocol.

Filecoin[10] **and Helium** demonstrate decentralized-network plays with token-incentive mechanisms. Both attracted speculative participants more than utility-driven users, primarily because the hardware in each case had no standalone consumer value. The purchase rationale collapsed if the token economics did not deliver expected returns. The reference Seed device — the Origin, built by Wylloh as the first commercial implementation of LMAP — has a primary consumer utility: the operator's film library, accessible to LMAP-compatible playback clients on

their local network. This utility does not depend on any token-incentive mechanism. Network participation is a beneficial side effect of a device that operators would buy regardless. This sidesteps the chicken-and-egg adoption problem that has limited prior decentralized-CDN plays.

Bitcoin[11] **and Ethereum**[12] establish the cryptographic and economic primitives this protocol relies upon. The protocol's contributions specific to film distribution are the dual-contract architecture (§3), the modular rights-stacking mechanism (§4), the two-tier attestation model (§7), the permanence guarantee (§8), and the Seed-network distribution model (§9).

17. Conclusion

We have proposed a protocol for film distribution addressing fundamental misalignments in centralized streaming. By tokenizing films into standardized units with embedded licensing rights, we create genuine digital ownership. By distributing content through a peer-to-peer network growing stronger with demand, we invert economics that punish platforms for success. By automating royalty distribution including on secondary sales, we ensure transparent creator compensation through markets that never existed for physical media.

The two-tier attestation model resolves the perceived conflict between open-protocol architecture and contractual studio trust requirements. The open tier accepts permissive content with intentional cryptographic permeability, in exchange for the platform-independence guarantee that makes the protocol durable. The certified tier provides hardware-attested per-device key wrapping for content where contractual DRM is required, governed by a federated attestation framework structurally independent from any commercial operator.

The separation of distribution tokens from copyright ownership enables fluid rights markets while preserving creator control over intellectual property. Commercial exhibitors can acquire rights directly from collectors. Studios can acquire copyright for derivative works without disrupting existing token markets. Rights flow to natural owners through market mechanisms rather than negotiated deals.

The protocol requires no permission from existing gatekeepers. Public-domain content bootstraps the network, demonstrating capabilities before commercial titles join. As the network grows, established platforms face a choice: implement the protocol and benefit from its efficiencies, or cede ground to peer marketplaces and competitors that do.

LMAP represents not disruption but recoherence — using peer-to-peer technology once deployed to undermine the industry to instead restore fair compensation and sustainable economics. The same distributed networks that enabled unauthorized copying can now enable legitimate, liquid, permanent ownership. *In the venom, the antidote.*

Implementation Status Summary

A precise summary of the protocol's components by their status as of this paper's publication:

- **Shipped (legacy V4.1 deployment):** ERC-1155 distribution registry on Polygon mainnet at `0x8e834c6031A2d59e1cADd24Ab24F668301ed7c1D`; role-gated minting; modular rights stacking; chunked AES-256-GCM encryption; open-tier deterministic key derivation; self-contained manifest reading from on-chain metadata + IPFS, with storage-service fallback; open-source reference Seed daemon (verify ownership, fetch + decrypt, auto-pin, LAN-stream over HTTP/mDNS); web client at wylloh.com; first tokenized film (*The Cocoanuts*, 1929) sold and downloaded end-to-end across two independent architectures (macOS arm64, Raspberry Pi aarch64) with byte-identical output. The V4.1 contract continues to serve existing tokens.
- **Specified, in active implementation (V5):** the LMAPRegistryV5 reference contract under foundation governance — adds the three-way revenue split (§10.1), publisher/author distinction (§10.2), royalty shareholder distribution (§10.3), atomic batch acquisition for rights stacking (§10.5), `mediaType` field for media-agnostic operation, updateable platform treasury via multi-sig timelock, ERC-4337 paymaster hooks, and permissionless

minting (§15). Scaffold with passing tests; full marketplace implementation, comprehensive test coverage, and deployment under foundation governance are 2026 milestones.

- **Specified, in implementation (hardware and clients):** reference Seed device (the *Origin*); native client applications for Roku, Apple TV, and iOS.
- **Specified, future protocol versions (V6+):** ERC-721 copyright registry; on-chain staking for commercial-exhibition windows; presale-funding milestone-escrow contracts; threshold dispersal of ciphertext shards; certified-tier attestation framework with federated issuers and **normatively-required hardware-attested per-device key wrapping** (§7.2). The wire-level attestation specification (report format, credential schema, issuer federation protocol, revocation flow) is committed to a separate LMAP V6 technical specification, forthcoming.
- **Long-horizon (operational maturity required):** studio-grade certified-tier engagement with major rights holders; native protocol token mechanics for proof-of-storage incentives.

This paper is a working specification. Components shipped today behave as described; components specified are commitments the protocol is engineered toward. The Liquid Media Foundation, when formally established, will maintain this status summary as a living public artifact.

Changelog

Version 2.3 (May 2026). Reference Seed daemon promoted from "specified" to "shipped" in the Implementation Status. §9 (Seed Network) updated to reflect end-to-end operation of the open-source reference daemon on independent ARM64 hardware (Raspberry Pi 4) with byte-identical decrypted output across architectures and no dependency on any centralized service. The daemon's operating behavior — IPFS fetch with local Kubo preference, automatic content pinning to make the Seed a provider, LAN-streaming via HTTP/mDNS with byte-range support — is documented as the canonical reference behavior for compliant Seeds. §7.2 (Certified Tier) gains a normative statement: hardware binding (SE-resident keypair generation, non-extractable private keys, in-SE key unwrapping, in-SE attestation signing) is a MUST for V6 conformance, and an implementation omitting any of these operations is not LMAP-V6-conformant and MUST NOT be marketed as such. A clarifying paragraph distinguishes *playback* (which requires certified hardware) from *carriage* (which does not — tokens, encrypted content, and pinning remain freely accessible across all hardware), mirroring DCI theatrical distribution. Wire-level attestation specifics (report format, credential schema, issuer federation protocol, revocation flow) are deferred to a separate LMAP V6 technical specification, forthcoming.

Version 2.2 (May 2026). V5 reference contract specification additions. §10 (Marketplace Mechanics) substantially expanded to specify the V5 economic model: three-way revenue split (protocol 2.5% / publisher 0–25% / author remainder), distinct publisher/author identities (publisher cryptographically authentic via mint-time `msg.sender`), royalty shareholder distribution supporting up to 50 collaborators per title, atomic batch acquisition for rights stacking, and an honest articulation of the royalty enforcement boundary (V5-marketplace sales enforce cryptographically; direct ERC-1155 transfers preserve the gift / inheritance use case). §15 (Governance) updated to reflect V5's permissionless minting model — curation moves from a protocol-level role gate to a three-signal application-layer model: cryptographic publisher identity, a foundation-maintained verified publishers registry at `lmap.org/publishers.json`, and individual platforms' editorial choice. Implementation Status restructured to reflect V4.1 (legacy, in operation) and V5 (in active implementation).

Version 2.1 (May 2026). Rebrand pass under Liquid Media Foundation stewardship. The protocol — previously referred to internally as "Wylloh" — is renamed *LMAP* (Liquid Media Access Protocol) to formalize the separation between the open standard (LMAP, stewarded by the foundation) and the first commercial implementation (Wylloh, the company building the reference hardware and storefront). Substantive technical content of v2.0 is preserved

unchanged. References to commercial-platform UX details that exceed protocol scope have been generalized. A more substantive v3.0 with deepened §15 (foundation governance) and expanded §16 (comparison with prior work) is forthcoming.

Version 2.0 (April 2026). Comprehensive revision following the April 2026 mainnet launch and the architectural pivot to the headless-Seed network. Key additions: §5 (layered architecture), §7 (two-tier attestation model with attestation issuers and the federated framework), §8 (permanence guarantees), §15 (governance and the foundation), §16 (comparison with prior work). The cryptographic claims of v1's §6 have been rewritten in v2's §7 to reflect the actual two-tier model rather than the threshold-key-sharing scheme v1 described aspirationally. The original V1 PDF is preserved in the project archive.

Version 1.0 (April 2026). Initial publication.

References

1. W. Radomski et al., "EIP-1155: Multi Token Standard," Ethereum Improvement Proposals, 2018.
2. W. Entriken et al., "EIP-721: Non-Fungible Token Standard," Ethereum Improvement Proposals, 2018.
3. J. Benet, "IPFS — Content Addressed, Versioned, P2P File System," arXiv:1407.3561, 2014.
4. A. Shamir, "How to Share a Secret," *Communications of the ACM*, vol. 22, no. 11, pp. 612–613, 1979.
5. ARM Limited, "ARM TrustZone Technology," developer.arm.com, current edition.
6. Trusted Computing Group, "TPM 2.0 Library Specification," trustedcomputinggroup.org, current edition.
7. H. Birkholz et al., "Remote ATtestation procedureS (RATS) Architecture," IETF RFC 9334, January 2023.
8. H. Krawczyk and P. Eronen, "HKDF: HMAC-based Extract-and- Expand Key Derivation Function," IETF RFC 5869, May 2010.
9. V. Buterin et al., "ERC-4337: Account Abstraction Using Alt Mempool," Ethereum Improvement Proposals, 2021.
10. Protocol Labs, "Filecoin: A Decentralized Storage Network," filecoin.io technical specification.
11. S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.
12. V. Buterin, "Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform," 2014.